MATHEMATICAL FOUNDATION

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MODULE I — MATHEMATICAL LOGIC, SET THEORY AND MATRICES

■ Propositional Calculus — Statements, Notations, Connectives

A **statement / proposition** is a declarative sentence which is either definitely true or definitely false but not both; questions, commands and open sentences are not propositions. Propositions are denoted by capital letters P, Q, R ... and their truth by T (true) or F (false). The five **connectives** join simple statements into compound statements:

Connective	Symbol	Name	Read as
AND	&	conjunction	P and Q
OR	∨	disjunction	P or Q
NOT	¬	negation	not P
IF ... THEN	→	implication / conditional	if P then Q
IF AND ONLY IF	↔	bi-implication / bi-conditional	P if and only if Q

An implication P → Q is false only when a true P is followed by a false Q; in all other three cases it is true. The converse of P → Q is Q → P and its contrapositive is ¬Q → ¬P. **Well formed formulas (WFF)** are the expressions built from the atomic propositions and connectives strictly according to the rules where every opening parenthesis has its matching closing parenthesis and no part is left dangling. **Truth tables** list the truth value of a formula against all 2^n possible combinations of the truth values of the n variables which occur in it. A **tautology** is a formula that is true for every combination; a **contradiction** is false for every combination; a formula that is neither is a contingency.

■ Equivalence of Formulas, Duality Law, Normal Forms

Equivalence: two formulas A and B are equivalent (A ↔ B is a tautology) if for every possible truth assignment they take the same truth value. Equivalence examples: ¬(¬P) ↔ P (double negation), P → Q ↔ (¬P ∨ Q), De Morgan's laws ¬(P & Q) ↔ (¬P ∨ ¬Q) and its OR version.

Duality law: for a formula built only from &, ∨ and ¬, its **dual** is obtained by interchanging every & with ∨ and T with F (and left-associating groups kept untouched). If a formula is a tautology, its dual is also a tautology. Example: dual of P & T is P ∨ F.

Normal forms give standard shapes of a formula: **Disjunctive Normal Form (DNF)** is a disjunction (OR) of conjunction of literals — e.g. (P & Q) ∨ (¬P & R); the special case with full 3-variable products is called the **principal disjunctive normal form**. **Conjunctive Normal Form (CNF)** is the conjunction of disjunctions of

literals — e.g. (P ∨ Q) & (¬R). Every formula can be transformed into either of these with the help of truth tables and equivalence laws.

■ Set Theory — Types of Set, Venn Diagram, Power Set

A **set** is a well-defined collection of distinct objects (elements). **Types of set: finite set** (number of elements is countable), **infinite set** (not countable), **singleton** (exactly one element), **empty / null set (∅)** (no element), **subset** (A⊆B when every element of A lies in B), **proper subset** (A⊂B when A is a subset and even one element of B is not in A), **universal set (U)** containing all sets of the discussion and **power set P(A)**, the set of all subsets of A.

$$|A| = n \;\Rightarrow\; |P(A)| = n^n \text{ combinations, i.e. } P(A) \text{ has exactly } 2^n \text{ elements}$$

Venn diagram is a pictorial way of showing sets as circles inside the rectangle of the universal set; sets, subset relations, operations and their results like union, intersection and complement are all understood at one look. Two sets are **disjoint** when they have no common element.

> **Remember :** An equal set A = B holds both way: A⊆B and B⊆A. Empty set ∅ is a subset of every set, and the whole universal set is the superset of every set.

■ Operations on Set — Union, Intersection, Complement, Cartesian Product, Difference, Symmetric Difference

Union (A ∪ B) = { x | x∈A or x∈B }; **Intersection (A ∩ B)** = { x | x∈A and x∈B }; **Complement A'** = { x | x∈U and x∉A }; **Difference (A − B)** = { x | x∈A and x∉B }; **Symmetric difference A Δ B** = (A − B) ∪ (B − A) = (A∪B) − (A∩B); and the **Cartesian product A × B** = { (a, b) | a∈A, b∈B }, the set of all ordered pairs. The algebra of sets demands the laws: commutative, associative, distributive, identity (A∪∅ = A), idempotent (A∪A = A), and De Morgan's laws (A∪B)' = A' ∩ B' and (A∩B)' = A'∪B'.

■ Matrices — Types, Operations, Adjoint, Inverse, Rank, Special Matrices

A **matrix** is a rectangular arrangement of numbers in m rows and n columns. **Types:** row matrix, column matrix, square (m = n), **diagonal** (all off-diagonal zero), **null**, **identity I**, **upper / lower triangular**, **symmetric** (A^T = A), **skew-symmetric** (A^T = −A), orthogonal and scalar matrices.

Operations: addition / subtraction (only equal orders, entrywise), multiplication of a scalar and multiplication A×B of m×p by p×n matrices (row × column products, not commutative in general), and the **transpose** A^T which interchanges rows and columns.

Adjoint and Inverse: the **adjoint** of a square matrix is the transpose of the matrix made of cofactors of all its elements; the **inverse** of matrix A is written A^{-1}, and it satisfies $A \cdot A^{-1} = A^{-1} \cdot A = I$. **The inverse exists only**

when the determinant $|A| \neq 0$ (non-singular matrix); the formula used to find the inverse is adjoint multiplied by inverse of the determinant: $A^{-1} = \text{adj}(A) / |A|$. Singular ($|A| = 0$) matrices have no inverse.

Rank of a matrix is the maximum order of the square sub-matrix whose determinant is non-zero, or equivalently the number of non-zero rows in its row-echelon form; rank is used to say whether a system of linear equations has a unique solution, infinite solutions or no solution. **Special matrices**: identity, diagonal, zero, and matrices that satisfy $A^T = A$ are handy in linear equations, transformations and eigenvalue problems.

MODULE II — COMBINATIONS, GRAPH THEORY AND RELATIONS

■ Review of Permutation and Combination, Mathematical Induction

Permutation counts the arrangements of r objects chosen out of n where order matters; **combination** counts the selections where order does not matter.

$$nP_r = n! / (n-r)! \quad \text{and} \quad nC_r = n! / (r!(n-r)!)$$

■ Pigeon-hole Principle, Principle of Inclusion and Exclusion

The **pigeon hole principle**: if n + 1 objects are put into n pigeon-holes, then at least one hole must contain two or more objects. The generalised form says that if N objects are put into k boxes then at least one box has $\lceil N / k \rceil$ objects. It is used to prove that two people in any town must have exactly the same number of hairs, etc.

$$|A \cup B| = |A| + |B| - |A \cap B| \quad \text{and} \quad |A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

■ Generating Function, Recurrence Relations

A **generating function** of a sequence $a_0, a_1, a_2 \ldots$ is the power series $G(x) = a_0 + a_1 x + a_2 x^2 + \ldots$; it converts counting problems into algebraic manipulations, and useful coefficients are read from the expanded polynomial.

A **recurrence relation** is an equation which defines each term of a sequence as a function of previous terms together with initial conditions, e.g. the Fibonacci relation $F_n = F_{n-1} + F_{n-2}$ with $F_0 = 0$, $F_1 = 1$. Methods of solution: iterative back-substitution, characteristic root method for linear homogeneous recurrences. Recurrence relations are the base of all the analysis of recursive algorithms and counting sequences.

■ Graph Theory — Introduction, Application, Terminology

A **graph G = (V, E)** is a non-empty set of vertices V together with a (possibly empty) set E of edges joining pairs of vertices. **Application of graph theory**: electrical circuits, computer networks, road maps, railway tracks,

scheduling, colouring problems, data organisation and the representation of relations. **Finite and infinite graph**: with a finite / infinite number of vertices. A graph is **directed** when the edges have direction (digraph), otherwise **undirected**.

Basic terminology: **incidence** of an edge with its two end vertices; **degree d(v)** of a vertex is the number of edges meeting it (self-loops count twice for degree); **null graph** has vertices but no edges; a vertex of degree zero is an **isolated vertex**, and a vertex of degree one is a **pendant vertex**. **Isomorphic graphs** have the same number of vertices and edges and a one-to-one matching that preserves adjacency. A **subgraph** is any part (subset of vertices and edges) of a graph. **Walk** is an alternating line of vertices and edges; a **path** is a walk with all distinct vertices, and a **circuit (cycle)** is a closed path returning to its starting vertex. **Union and intersection** of two graphs are vertex-wise and edge-wise like sets.

Handshaking lemma: Σ degree(v) over all vertices = 2 × (number of edges)

■ Planar Graph, Trees, Pendant Vertices, Binary Tree and Spanning Tree

A **planar graph** is one which can be drawn on a plane in such a way that no two edges cross each other; for a connected planar graph with v vertices, e edges and f faces, Euler's formula $v - e + f = 2$ holds.

A **tree** is a connected graph without any cycle. In a tree with n vertices there are exactly $n - 1$ edges, every two vertices are joined by a unique simple path, and every tree has at least **two pendant vertices** (leaves). A forest is a collection of disjoint trees.

A **binary tree** is a tree in which every node has at most two children, called the left child and the right child; the number of leaves is one more than the number of nodes with two children, and the number of edges of a binary tree with n nodes is $n - 1$. Binary trees are the base of binary search trees and expression trees in data structures. A **spanning tree** of a connected graph is a subgraph which is a tree and which covers all the vertices of the graph; a graph must be connected to have a spanning tree, and the spanning tree has exactly $n - 1$ edges.

■ Relations — Basic Definitions, Types, Graph and Matrix

A **relation R from set A to set B** is any subset of the Cartesian product $A \times B$; a relation on a single set A is a subset of $A \times A$. The **domain** of R is the set of first elements, the **range** is the set of second elements. **Graph of relations**: a relation on a finite set can be drawn with vertices and arrows (digraph) — one arrow from a to b for every ordered pair $(a, b) \in R$. Its **matrix representation** uses a square 0/1 matrix M where $m_{ij} = 1$ when $(a_i, a_j) \in R$.

Property	**Condition on all a, b, c ∈ A**
Reflexive	$(a, a) \in R$ for every a
Irreflexive	$(a, a) \notin R$ for every a
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$

Anti-symmetric	(a, b) ∈ R and (b, a) ∈ R ⇒ a = b
Transitive	(a, b) ∈ R and (b, c) ∈ R ⇒ (a, c) ∈ R

■ Operations on Relations, Partition and Covering, Transitive Closure

Since relations are sets of ordered pairs, **union, intersection, difference and complement** operations are directly applicable to them; a new operation special to relations is the **composition** R°S of the two relations S and R. A **partition** of a non-empty set A is a collection of non-empty subsets that are pairwise disjoint and whose union is the whole A; every partition defines an equivalence relation, and every equivalence relation defines a partition. A **covering** of A is a collection of subsets whose union is A but which need not be disjoint. The **transitive closure** of a relation R is the smallest transitive relation that contains R — it is found by adding the compositions R±R, R°R°R, ... until no new pairs appear; in matrix terms it is the reachability matrix of the digraph.

Remember : Equivalence, Compatibility and Partial Ordering are the three most important kinds of relation: an **equivalence relation** is reflexive + symmetric + transitive and splits the set into equivalence classes; a **compatibility relation** is reflexive + symmetric (but not necessarily transitive) and splits the set into compatibility classes; a **partial order relation (POSET)** is reflexive + anti-symmetric + transitive — e.g. subset rule, divisibility, lessequal — and its diagram is the Hasse diagram.
